1.0 UNDERSTAND INFORMATION SYSTEM.

1.1 Define information system.

Information system: an integrated set of components for collecting, storing, and processing data and for providing information, knowledge, and digital products. Business firms and other organizations rely on information systems to carry out and manage their operations, interact with their customers and suppliers, and compete in the marketplace. Information systems are used to run inter-organizational supply chains and electronic markets. For instance, corporations use information systems to process financial accounts, to manage their human resources, and to reach their potential customers with online promotions. Many major companies are built entirely around information systems.

1.2 Explain information system.

Many who come across the term information system for the first time think of it as some software based on information storage or something like that. Well, the name does sound that way. However, an information system is way bigger than that. So, what is an information system?

An information system is a combination of software, hardware, and telecommunication networks to collect useful data, especially in an organization. Many businesses use information technology to complete and manage their operations, interact with their consumers, and stay ahead of their competition. Some companies today are completely built on **information technology**, like eBay, Amazon, Ali Baba, and Google

Typical components of information systems

Now that you know what an information system is, let's look at its components. It has five components-hardware, software, data, and telecommunications.

- **Hardware:** This is the physical component of the technology. It includes computers, hard disks, keyboards, iPads, etc. The hardware cost has decreased rapidly while its speed and storage capacity has increased significantly. However, the impact of the use of hardware on the environment is a huge concern today. Nowadays, storage services are offered from the cloud, which can be accessed from telecommunications networks.
- **Software:** Software can be of two types, system software and application software. The system software is an operating system that manages the hardware, program files, and other resources while offering the user to control the PC using GUI. Application software is designed to manage particular tasks by the users. In short, system software makes the hardware usable while application software handles specific tasks. An example of system software is Microsoft windows, and an example of application software is Microsoft Excel. Large companies may use licensed applications which are developed and managed by software development companies to handle their specific needs. The software can be proprietary and open source, available on the web for free use.
- **Data:** Data is a collection of facts and is useless by themselves, but when collected and organised together, it can be very powerful for business operations. Businesses collect all the data and use it to make decisions that can be analysed for the effectiveness of the business operations.
- **Telecommunications:** Telecommunication is used to connect with the computer system or other devices to disseminate information. The network can be established

using wired or wireless modes. Wired technologies include fiber optics and coaxial cable, while wireless technologies include radio waves and microwaves.

Examples of information systems

Information systems have gained immense popularity in business operations over the years. The future of information systems and their importance depends on automation and the implemenration of AI technology.

Information technology can be used for specialised and generalised purposes. A generalised information system provides a general service like a database management system where software helps organise the general form of data. For example, various data sets are obtained using a formula, providing insights into the buying trends in a certain time frame.

On the contrary, a specialised information system is built to perform a specific function for a business. For example, an expert system that solves complex problems. These problems are focused on a specific area of study like the medical system. The main aim is to offer faster and more accurate service than an individual might be able to do on his own.

Facts of information systems

The products of information technology are part of our daily lives. Here are some of the facts about information systems.

- **Necessary for businesses to grow:** Every organization has computer-related operations that are critical to getting the job done. In a business, there may be a need for computer software, implementation of network architecture to achieve the company's objectives or designing apps, websites, or games. So, any company that is looking to secure its future needs to integrate a well-designed information system.
- **Better data storage and access:** Such a system is also useful for storing operational data, documents, communication records, and histories. As manual data may cost a lot of time, information systems can be very helpful in it. Information system stores data in a sophisticated manner, making the process of finding the data much easier.
- **Better decision making:** Information system helps a business in its decision-making process. With an information system, delivering all the important information is easier to make better decisions. In addition, an information system allows employees to communicate effectively. As the documents are stored in folders. it is easier to share and access them with the employees.

1.3 Identify types of information system.

Information systems are an integral part of every organization especially for maintaining cyber security. There are different types of information systems in cyber security and each of them is equally useful in companies. The different types of information systems with examples are enumerated as follows:

- **Transaction Processing System (TPS):** A transaction processing system is used to automate the process of collection, modification, and retrieval. The distinctive characteristics of this system are that it increases the reliability, continuity, and consistency of various business transactions and increases business performance. It helps companies to improve their daily performance and run their day-to-day operations hassle-free.

You will understand the applications of various information systems once you are well-versed in the types of information systems. The transaction processing system makes the collection, modification, and retrieval process very flexible and accurate so that the possibility of errors is diminished.

- **Office Automation System (OAS):** An office automation system is used for automating various administrative processes that are involved in an office. Automatic process of office transactions documenting and recording data is very important so that the possibility of errors is reduced and proper recording of transactions takes place. It is a part of the administrative process in an office and the office automation system helps it easier for the supervisors to look into these particular matters.

  The different kinds of information systems focus on improving every aspect of an organization and the office automation system is one that focuses on the administrative process of a company. It is divided into clerical activities and managerial jobs and the business activities that are done under this information system are stated as follows:

  - Word processing
  - Voicemail
  - E-mail

- **Knowledge Work System (KWS):** A continuous flow of updated and new versions of knowledge is very important in every company. To maintain that all the organizations have different knowledge work systems. A knowledge work system integrates the knowledge and the work system of an organization so that a company can successfully work in a manner that is integrated with knowledge.

  A knowledge work system also provides a lot of support to various professionals in a company so that new knowledge techniques and strategies can be implemented. These techniques may include group collaboration systems and artificial intelligence applications for sharing knowledge.

  The following are the aspects of a knowledge work system:

  - Computer-aided design systems are used to automate the designing process.
  - Huge amounts of financial data are analyzed by the financial workstations using the knowledge work system.
  - Graphics and virtual reality systems are used in knowledge work systems to present data.

- **Management Information System (MIS):** Management information system helps a lot in favor of the managers as it automates many processes that are previously done with the help of manual or human resources. Various business activities such as tracking business performance, defining business workflow, making a business plan, decision making, and analysis are now done with the assistance of management information systems. It also provides accurate feedback to the managers after analyzing the roles and responsibilities of the team members.

  A management information system is considered an important tool for managers because of the following reasons:

  - A proper report of the organizational performance is generated.
  - It increases the efficiency and productivity of the organization.
  - It thrives on improvement and innovation.
  - It provides assistance for planning and communication in business processes.
  - The organization gains a competitive advantage.

- **Decision Support System (DSS):** A decision support system helps to analyze the data and business performance so that proper and accurate decisions can be taken for the improvement of the business performance. It helps to automate the decision-making and problem-solving process of the company by analyzing the data and other information-related areas. A decision support system is also used by the managers in the time of advertising their business.

  Generally, the decision support system analyzes the sales, revenue, and inventory of a company and generates reports accordingly. It is a famous information system used across many industries and companies to achieve efficiency and accuracy.
- **Executive Support System (ESS):** An executive support system is generally used by the stakeholders of a company. The top-level executives use this information system for controlling the workflow and supervising the operations of the whole company. They also make important business decisions so that the company may grow further.

  An executive support system has the following unique characteristics:
  - It provides better telecommunication opportunities.
  - The executives receive effective display options and better computing capabilities.
  - The reports are based on textual information, static reports, and graphs.
  - It helps to monitor the performances and direct the course of action.
  - It looks into the competitor's strategies and makes forecasts for future trends.

1.4 Explain computer information system.

Computer information systems (CIS) are used to collect, organize, and distribute data. They are crucial for organizations managing large amounts of data electronically. Familiarity with information systems is a valuable skill in all industries due to the high demand for professionals who can develop data management solutions. Many companies rely heavily on information systems to stand out against competitors. You can use the following article to learn more about this field and what you can do with a degree in computer information systems.

What are computer information systems?

Computer information systems is a broad term. It refers to managing communications between hardware and software on data storage and management devices. Examples of these devices include databases, cloud storage, and servers. Computer information system architecture depends on the data being collected and how an organization plans to use it. Computer information systems professionals (like systems managers and database administrators) determine the most efficient setup and handle hardware/software integrations.

What is the difference between computer information systems and computer science?

Computer science more broadly deals with designing systems or software that can solve a particular issue. In contrast, computer information systems focus on how to use that software in various contexts.